WORKED SOLUTION

STEP II 2012, Question 11

Mechanics • Inelastic Collisions

STEP II 2012, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

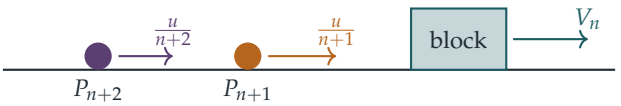
step.maths.org/questions/12-s2-q11

The key idea

Look at the momentum each particle carries: mass im times speed u/i is mu, the same for every particle. So after n particles have been absorbed the block's momentum is simply nmu, and conservation of momentum gives its speed at once. Whether the next particle ever arrives is then a comparison of two speeds, and the energy loss is "kinetic energy in minus kinetic energy out".

Solution

We model the block and the particles as particles moving along one straight line on a smooth surface, so no horizontal external force acts and momentum along the line is conserved in every collision. Each collision is perfectly inelastic, since the particle embeds in the block.



The speed after the nth collision

Particle P_i has momentum $im \cdot \frac{u}{i} = mu$, independent of i.

Suppose the nth particle collides with the block. Then particles $P_1, \ldots, P_{n-1}$ must have collided before it: the particles are fired in order and each is slower than the one before (u/i decreases), so a later particle can never overtake an earlier one, and no particle can pass the block without hitting it. So just after the nth collision the block contains $P_1, \ldots, P_n$, and its mass is

$$km + m(1 + 2 + \cdots + n) = m\left(k + \tfrac{1}{2}n(n+1)\right).$$

The block started at rest, and each absorbed particle brought momentum mu, so by

conservation of momentum the total momentum is now nmu. Hence the speed V_n of the block after the nth collision satisfies

$$m\Big(k+\tfrac{1}{2}n(n+1)\Big)V_n = nmu \quad\Longrightarrow\quad V_n = \frac{2nu}{2k+n(n+1)},$$

as required.

The number of collisions when $2k = N(N+1)$

After n collisions the block moves at the constant speed V_n (the surface is smooth). Particle P_{n+1}, which is behind the block, catches it if and only if it is strictly faster:

$$\frac{u}{n+1} > \frac{2nu}{2k+n(n+1)} \iff 2k+n(n+1) > 2n(n+1) \iff 2k > n(n+1),$$

where we multiplied up by the positive quantities $n+1$ and $2k+n(n+1)$. (For $n=0$ this reads $2k>0$, which is true: P_1 certainly hits the block at rest.)

> **?** **Ask yourself** Once one particle misses, could a later one still hit the block?

No. If P_{n+1} fails to catch the block, the block keeps the speed $V_n \geq \frac{u}{n+1}$ for ever, and every later particle is slower still, since $\frac{u}{i} < \frac{u}{n+1}$ for $i > n+1$. So the collisions are exactly $P_1, P_2, \ldots$ up to the first particle that fails.

Now take $2k = N(N+1)$. The condition for P_{n+1} to hit becomes $n(n+1) < N(N+1)$, that is $n < N$, because $n(n+1)$ is strictly increasing in $n \geq 0$. So $P_1, \ldots, P_N$ all hit the block, and P_{N+1} does not: with $n = N$ we have equality, and indeed

$$V_N = \frac{2Nu}{N(N+1)+N(N+1)} = \frac{u}{N+1},$$

exactly the speed of P_{N+1}, which therefore stays a fixed distance behind the block. Hence there are exactly N collisions.

The kinetic energy lost

Only $P_1, \ldots, P_N$ take part in collisions, so the total loss is the kinetic energy of those particles at firing (the block starts at rest) minus the final kinetic energy of the block.

Initially,

$$\sum_{i=1}^{N} \tfrac{1}{2}(im)\Big(\frac{u}{i}\Big)^2 = \tfrac{1}{2}mu^2 \sum_{i=1}^{N} \frac{1}{i}.$$

Finally the block has mass $m\big(k+\frac{1}{2}N(N+1)\big) = mN(N+1)$ and speed $\frac{u}{N+1}$, so its kinetic energy is

$$\tfrac{1}{2}\,mN(N+1)\,\frac{u^2}{(N+1)^2} = \tfrac{1}{2}mu^2\,\frac{N}{N+1} = \tfrac{1}{2}mu^2\Big(1-\frac{1}{N+1}\Big).$$

Subtracting,

$$\begin{aligned}\text{KE lost} &= \tfrac{1}{2}mu^2\left(\sum_{i=1}^{N}\frac{1}{i} - 1 + \frac{1}{N+1}\right)\\ &= \tfrac{1}{2}mu^2\left(\frac{1}{2}+\frac{1}{3}+\cdots+\frac{1}{N}+\frac{1}{N+1}\right) = \tfrac{1}{2}mu^2\sum_{n=2}^{N+1}\frac{1}{n},\end{aligned}$$

which is the required result.

Checks

- $N = 1$ ($k = 1$): one particle of mass m at speed u hits a block of mass m, giving speed $\frac{1}{2}u$ and loss $\frac{1}{2}mu^2 - \frac{1}{4}mu^2 = \frac{1}{4}mu^2$, matching $\frac{1}{2}mu^2 \cdot \frac{1}{2}$. Then P_2 also moves at $\frac{1}{2}u$ and never arrives.
- $N = 2$ ($k = 3$): the formula gives loss $\frac{1}{2}mu^2(\frac{1}{2}+\frac{1}{3}) = \frac{5}{12}mu^2 \approx 0.4167\,mu^2$. An exact step-by-step simulation of the motion (positions and collision times in fractions) gave the same value, exactly two collisions, and speeds V_n matching the formula; it also confirmed N collisions and the loss formula for every N up to 11.
- Units: the loss is a mass times a speed squared, as an energy should be.

Common slips

- Using only the nth particle's momentum and forgetting that the block already contains $P_1, \ldots, P_{n-1}$ (or forgetting the block's own mass km).
- Counting $N + 1$ collisions: P_{N+1} has exactly the block's speed, so it never makes contact.
- Including the kinetic energy of $P_{N+1}, P_{N+2}, \ldots$ in the loss; they never collide, so they lose nothing.

Beyond STEP

The block is a body whose mass grows as it sweeps up material, a cousin of the rocket and the raindrop falling through a cloud. For such variable-mass systems one applies Newton's second law in the form "rate of change of momentum equals force" to the whole system, exactly as we did collision by collision. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats variable-mass systems and rocket motion carefully.

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